

PARANA, Argentina

WMO: 873740

Lat: 31.7903S

Lon: 60.4842W

Elev: 78

StdP: 100.39

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.7	4.0	-3.6	2.8	7.7	-2.0	3.2	8.3	11.6	12.9	10.3	13.6	1.5	180	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	34.4	23.2	33.0	22.8	31.8	22.4	25.7	31.4	24.8	30.3	24.0	29.3	4.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.1	29.1	23.2	18.1	28.1	22.3	17.2	27.1	79.8	31.4	76.1	30.4	72.7	29.3	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.3	8.2	DB	0.2	37.1	1.3	1.1	-0.8	37.9	-1.5	38.6	-2.3	39.2	-3.2	40.0	
			WB	-1.2	27.4	1.4	1.0	-2.2	28.1	-3.0	28.6	-3.8	29.2	-4.8	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	25.3	24.0	21.9	18.7	15.4	12.8	12.2	13.8	15.9	18.8	21.5	23.8
	DBStd	5.63	2.73	2.85	3.01	3.51	3.55	3.73	4.08	4.28	3.89	3.40	3.13	2.99
	HDD10.0	47	0	0	0	0	2	12	21	11	2	0	0	0
	HDD18.3	811	0	1	6	37	106	173	197	154	94	34	8	1
	CDD10.0	3203	475	392	368	261	169	96	89	128	178	273	345	428
	CDD18.3	924	217	159	116	48	15	6	6	13	19	49	103	171
	CDH23.3	8213	2256	1407	887	288	53	12	22	94	176	403	928	1687
	CDH26.7	3078	1008	551	279	58	7	1	2	19	46	105	310	691
Wind	WSAvg	3.3	3.0	3.0	2.8	2.9	2.9	3.2	3.3	3.6	3.8	4.0	3.6	3.4
Precipitation	PrecAvg	1073	133	123	151	120	63	35	34	37	59	113	116	127
	PrecMax	1546	983	383	539	393	232	117	87	169	247	327	313	405
	PrecMin	529	10	4	24	5	0	0	0	0	3	16	7	16
	PrecStd	235	141	87	94	93	56	31	24	37	52	68	65	85
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	35.2	33.2	30.6	27.5	24.9	26.1	29.2	31.0	32.2	33.4	35.7
		MCWB	24.3	23.8	23.5	23.6	21.5	20.7	19.1	20.6	21.9	22.5	21.7	22.8
	2%	DB	34.7	33.2	31.3	28.4	24.8	22.4	23.1	26.0	27.8	29.3	31.5	33.7
		MCWB	23.2	23.6	22.9	21.7	19.9	19.2	18.2	18.8	19.4	20.5	21.0	22.6
	5%	DB	33.1	31.7	29.7	26.6	22.9	20.3	20.8	23.5	25.2	27.3	30.1	32.1
		MCWB	22.7	23.1	22.2	20.8	19.0	17.4	16.7	17.4	17.9	19.7	20.3	22.1
	10%	DB	31.8	30.2	28.1	24.8	21.0	18.6	18.7	21.0	22.9	25.5	28.3	30.5
		MCWB	22.5	22.5	21.2	19.8	17.6	15.7	15.3	15.9	16.1	18.4	19.7	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.7	25.8	24.7	22.4	21.3	20.3	21.6	23.0	24.4	24.4	26.5
		MCDB	32.4	31.4	31.4	29.0	26.1	24.0	24.1	27.5	29.0	30.4	30.2	32.6
	2%	WB	25.4	25.4	24.5	23.3	20.8	19.7	18.6	19.7	20.8	22.3	22.7	25.0
		MCDB	31.7	30.6	29.0	26.9	23.3	21.8	22.2	24.7	26.3	26.9	28.8	30.6
	5%	WB	24.5	24.4	23.3	21.9	19.6	18.2	17.3	18.4	18.9	20.9	21.7	23.7
		MCDB	30.6	29.4	27.5	25.1	22.1	20.2	20.2	22.6	23.3	25.4	27.2	29.3
	10%	WB	23.7	23.4	22.1	20.7	18.4	16.6	16.0	16.7	17.4	19.7	20.7	22.5
		MCDB	29.5	28.2	26.6	23.6	20.7	18.1	18.3	20.1	21.5	23.9	26.1	28.0
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.0	9.9	8.9	8.2	8.1	8.6	9.8	10.4	10.2	11.1	11.0
		MCDBR	12.6	12.0	11.7	11.0	9.7	9.0	10.0	12.2	13.4	12.7	13.2	12.7
		MCWBR	4.9	4.8	5.1	5.6	5.5	5.6	5.8	6.8	6.7	6.2	5.8	5.3
	5% WB	MCDBR	10.7	10.0	10.0	9.2	8.2	7.8	8.7	11.0	11.5	10.8	11.0	10.8
		MCWBR	5.2	4.9	5.5	5.4	5.2	5.5	5.7	6.7	6.7	6.3	5.8	5.6
Clear-Sky Solar Irradiance	taub	0.400	0.388	0.372	0.380	0.358	0.355	0.358	0.425	0.485	0.426	0.392	0.396	
	taud	2.384	2.433	2.468	2.413	2.436	2.435	2.397	2.159	1.997	2.240	2.363	2.378	
	Ebn at Noon	940	931	912	846	814	789	803	782	786	884	941	947	
	Edh at Noon	129	119	107	102	89	84	91	129	169	142	131	130	
All-Sky Solar Radiation	RadAvg	7.14	6.13	5.28	3.89	2.89	2.49	2.77	3.67	4.76	5.66	6.89	7.23	
	RadStd	0.47	0.47	0.44	0.50	0.25	0.27	0.29	0.33	0.41	0.47	0.40	0.42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	+0.23	+0.49	+0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page